Integral stochastic orders of m-generalized order statistics from transform-ordered nonparametric families

Idir Arab
Centro de Matemática da Universidade de Coimbra
Department of Mathematics
University of Coimbra, Portugal
e-mail: idir.bhh@gmail.com

Tommaso Lando
Department of Economics
University of Bergamo, Italy
email: tommaso.lando@unibg.it

Paulo Eduardo Oliveira
Centro de Matemática da Universidade de Coimbra
Department of Mathematics
University of Coimbra, Portugal
e-mail: paulo@mat.uc.pt

Tomasz Rychlik
Institute of Mathematics
Polish Academy of Sciences
Śniadeckich 8, 00656 Warsaw, Poland
e-mail: trychlik@impan.pl

Abstract

Mathematics Subject Classification 2020: 60E15, 62G30, 62G32.

Keywords and phrases: generalized order statistics, m-generalized order statistics, k-th record values, integral stochastic order, transform stochastic order.

1 Introduction

2 Preliminaries and main theorem

Generalized order statistics (see Kamps, 1995) is a semiparametric probability model based on a vector $(\gamma_1, \ldots, \gamma_n)$ or a sequence $(\gamma_1, \gamma_2, \ldots)$ of arbitrary positive parameters and a one-dimensional baseline distribution function F, say. This is defined in a two-stage procedure. A vector of random variables $(U_{1,\tilde{\gamma}_1}, \ldots, U_{n,\tilde{\gamma}_n})$ (sequence $(U_{1,\tilde{\gamma}_1}, U_{2,\tilde{\gamma}_2}, \ldots)$, respectively) is called a vector (sequence, respectively) of uniform generalized order statistics if it has a joint distribution identical with

$$U_{r,\tilde{\gamma}_r} \stackrel{d}{=} 1 - \prod_{i=1}^{r} U_i^{\frac{1}{\gamma_i}}, \qquad r = 1, \ldots, n \ (r = 1, 2, \ldots, \text{ respectively}), \tag{1}$$ {1}

where $U_1, \ldots, U_n$ ($U_1, U_2, \ldots$, respectively) are i.i.d. standard uniform random variables. We see that the distribution of the first r uniform generalized order statistics depends only on the first parameters $\gamma_1, \ldots, \gamma_r$. Accordingly, the subscript $(r, \tilde{\gamma}_r)$ in our notation means the number of generalized order statistics and respective vector of significant parameters $\tilde{\gamma}_r = (\gamma_1, \ldots, \gamma_r)$. The vector (sequence) of generalized order statistics $(X_{1,\tilde{\gamma}_1}, \ldots, X_{n,\tilde{\gamma}_n})$ ($(X_{\tilde{\gamma}_1}, X_{2,\tilde{\gamma}_2}, \ldots)$, respectively), with parameters $(\gamma_1, \ldots, \gamma_n)$ ($(\gamma_1, \gamma_2, \ldots)$, respectively), and a baseline distribution function F has the distribution as

$$X_{r,\tilde{\gamma}_r} \stackrel{d}{=} F^{-1}\left(1 - \prod_{i=1}^{r} U_i^{\frac{1}{\gamma_i}}\right), \qquad r = 1, \ldots, n \ (r = 1, 2, \ldots, \text{ respectively}). \tag{2}$$ {2}

According to (1) and (2), the distribution of first r generalized order statistics (gos) depends only on the first r parameters and the baseline distribution. Moreover, the marginal distribution of r-th generalized order statisticgos does not depend on the ordering of these parameters.

The most popular submodel of generalized order statisticsgos is the vector of standard order statistics based of n i.i.d. random variables with a marginal distribution function F. These are just generalized order statistics with parameters $\gamma_r = n + 1 - r$, $r = 1, \ldots, n$. Its generalization constitutes the model of progressively censored type II order statistics $X^{\mathbf{R}}_{1:n:N}, \ldots, X^{\mathbf{R}}_{n:n:N}$ with N objects under experiment, $n \leq N$ actual observations, and censoring plan $\mathbf{R} = (R_1, \ldots, R_n)$, where R_i is the number of randomly chosen objects removed from the experiment just after $X^{\mathbf{R}}_{i:n:N}$ so that $R_i \geq 0$ and $\sum_{i=1}^{n} R_i + n = N$. This undergoes the generalized order statisticsgos model with a finite sequence of parameters $\gamma_r = \sum_{i=r}^{n}(R_i + 1)$, $r = 1, \ldots, n$. Another classic example of generalized order statistics is the sequence of k-th upper records $R_1^{(k)}, R_2^{(k)}, \ldots$ with parameters $\gamma_i = k \in \mathbb{N}$ and a continuous baseline distribution function F. In particular, for $k = 1$ we obtain classic upper record values.

Marginal distributions of generalized order statistics, and in particular uniform ones are complicated in general, especially if some parameters (not all) are multiple. Recursive formulae were presented in Cramer and Kamps (2003). Explicit expressions appeared in Rychlik (2026). They simplify much in the case of so called m-generalized order statistics (m-gos) for which the parameters form arithmetic sequences. We can construct an infinite sequence of m-gos if the common difference is nonnegative. Otherwise the assumption of positivity of all gammas make possible constructing only a finite number of m-gos. The assumption of the m-gos model is satisfied by standard order statistics and k-th record values. The progressively censored type II order statistics satisfy this submodel if $R_2 = \cdots = R_n$, i.e., if we remove the same number of object after each failure except for possibly the first one. It follows that we can construct finite numbers of classic and progressively censored order statistics, and infinite numbers of record values.

3 Bounds for the probability of exceeding a gos

In this section, we extend the results of Section 7 in Arab et al. (2025). We are interested in the probability that the underlying random variable exceeds an expected gos, that is, $P(X \geq \mathbb{E}X_{r,\tilde{\gamma}_r})$.

The framework covers several practically relevant models. For example, in the case of regular order statistics, when the order statistic $X_{k:n}$ may represent a lifetime of a k-out-of-n system with i.i.d. components, one can establish bounds for the probability that the

lifetime of one component exceeds the expected lifetime of the system (see Arab et al., 2025). The same perspective applies to progressively censored type II order statistics; under a censoring scheme determined by the vector $\mathbf{R}$, $X^{\mathbf{R}}_{k:n:N}$ represents the time of the k-th failure in a test that starts with N items and stops after n failures ($n \leq N$). In this setting, we are interested in the probability that a single item lasts longer that the expected lifetime of the the k-th failure, according to the censoring plan $\mathbf{R}$. In the record value case, the problem reduces to studying the probabilty of exceeding the r-th record $R_r^{(k)}$ of order k, which is particularly natural when record values are used to model sequential extremes. Similar exceedance probabilities arise when $\mathbb{E}X_{r,\tilde{\gamma}_r}$ represents some threshold induced by some sampling plan.

Assuming that $G^{-1} \circ F$ is convex and applying Jensen's inequality we immediately find the bound

$$G^{-1} \circ F(\mathbb{E}X_{r,\tilde{\gamma}_r}) \leq \mathbb{E}G^{-1}\left(1 - \prod_{i=1}^{r} U_i^{1/\gamma_i}\right),$$

so that

$$P(X \leq \mathbb{E}X_{r,\tilde{\gamma}_r}) \leq G\left[\mathbb{E}G^{-1}\left(1 - \prod_{i=1}^{r} U_i^{1/\gamma_i}\right)\right].$$

The bound above depends just on G, r, and $\tilde{\gamma}_r$, the only assumption we need about F is whether it belongs to some convex-ordered familyand the baseline distribution F that needs to belong to some convex-ordered family of distributions. A similar argument applies to the case when $G^{-1} \circ F$ is concave. The results are summarized in the following proposition. These type of bounds been used to obtain convexity or concavity tests in Lando and Benjrada (2025).

Proposition 1. *Given some cumulative distribution function G, define*

$$\pi^{G}_{r,\tilde{\gamma}} = G\left[\mathbb{E}G^{-1}\left(1 - \prod_{i=1}^{r} U_i^{1/\gamma_i}\right)\right].$$

(a) If $G^{-1} \circ F$ is convex, $P(X \geq \mathbb{E}X_{r,\tilde{\gamma}_r}) \geq 1 - \pi^{G}_{r,\tilde{\gamma}}$.

(b) If If $G^{-1} \circ F$ is concave, $P(X \geq \mathbb{E}X_{r,\tilde{\gamma}_r}) \leq 1 - \pi^{G}_{r,\tilde{\gamma}}$.

The above bounds are useful in reliability but also have statistical applications, as they can be used to obtain tests for convexity, or concavity, of $G^{-1} \circ F$ (Lando and Benjrada, 2025).

Common choices of G yield explicit expressions for the bounds. For example, let $G = W_\alpha$. Since $W_\alpha^{-1}(u) = \frac{1-(1-u)^\alpha}{\alpha}$, using the independence of $U_1, \ldots, U_n$ and $\mathbb{E}(U_i^{\alpha/\gamma_i}) = \frac{1}{1+\alpha/\gamma_i}$, for every i, we obtain

$$\mathbb{E}\, W_\alpha^{-1}\left(1 - \prod_{i=1}^{r} U_i^{1/\gamma_i}\right) = \frac{1 - \mathbb{E}\left(\prod_{i=1}^{r} U_i^{\alpha/\gamma_i}\right)}{\alpha} = \frac{1 - \prod_{i=1}^{r} \mathbb{E}\left(U_i^{\alpha/\gamma_i}\right)}{\alpha} = \frac{1 - \prod_{i=1}^{r} \frac{\gamma_i}{\gamma_i+\alpha}}{\alpha}.$$

Plugging the above expression into W_α, we obtain

$$\pi^{W_\alpha}_{r,\tilde{\gamma}} = 1 - \left(\prod_{i=1}^{r} \frac{\gamma_i}{\gamma_i + \alpha}\right)^{1/\alpha}. \tag{3}$$ {explicit-pi-g

For the k-th record values (where $\gamma_i = k$), (3) becomes

$$\pi_{r,k}^{W_\alpha} = 1 - \left(\frac{k}{k+\alpha}\right)^{r/\alpha}. \tag{4}$$ {explicit-pi-r

The above expressions simplify further if we consider the most common special cases, for which $\alpha = -1, 1$ and $\alpha \to 0$.

The application of these bounds is straightforward. For example, assuming that F is IFR and DD, we have an upper and lower bound for $P(X \geq \mathbb{E}R_r)$, that is,

$$e^{-\frac{r}{k}} \leq P(X \geq \mathbb{E}R_r) \leq \left(\frac{k}{k+1}\right)^r.$$

If we weaken the conditions we have wider bounds. For example, assuming that F is IOR and DRFR, we now obtain

$$\left(\frac{k-1}{k}\right)^r \leq P(X \geq \mathbb{E}R_r) \leq \exp\left[-\sum_{i=1}^{\infty} \frac{1}{n}\left(\frac{k}{k+n}\right)^r\right].$$

The left bound above is trivial for $k = 1$.

References

I. Arab, T. Lando, and P. E. Oliveira. Inequalities and bounds for expected order statistics from transform-ordered families. *J. Appl. Probab.*, 62(3):1216–1234, 2025. doi: 10.1017/jpr.2024.121.

B. C. Arnold and H. Nagaraja. Lorenz ordering of exponential order statistics. *Statist. Probab. Letters*, 11(6):485–490, 1991. doi: 10.1016/0167-7152(91)90112-5.

Y. Chen, P. Embrechts, and R. Wang. Technical note—an unexpected stochastic dominance: Pareto distributions, dependence, and diversification. *Oper. Res.*, 73(3):1336–1344, 2025.

E. Cramer and U. Kamps. Marginal distributions of sequential and generalized order statistics. *Metrika*, 58(3):293–310, 2003. doi: 10.1007/s001840300268.

U. Kamps. A concept of generalized order statistics. *J. Statist. Plann. Inference*, 48(1): 1–23, 1995. doi: 10.1016/0378-3758(94)00147-N.

S. Kochar. Lorenz ordering of order statistics. *Statist. Probab. Letters*, 76(17):1855–1860, 2006. doi: 10.1016/j.spl.2006.04.032.

S. Kochar. Stochastic comparisons of order statistics and spacings: a review. *ISRN Probability and Statistics*, 2012. doi: 10.5402/2012/839473.

S. Kochar and M. Xu. Comparisons of parallel systems according to the convex transform order. *J. Appl. Probab.*, 46(2):342–352, 2009. doi: 10.1239/jap/1245676091.

V. Komornik. Another short proof of Descartes's rule of signs. *Amer. Math. Monthly*, 113 (9):829–830, 2006. doi: 10.2307/27642066.

A. Kundu and S. Chowdhury. Ordering properties of order statistics from heterogeneous exponentiated Weibull models. *Statist. Probab. Lett.*, 114:119–127, 2016. doi: 10.1016/j.spl.2016.03.017.

T. Lando and M. E.-S. Benjrada. A new class of tests for convex-ordered families based on expected order statistics. *Electron. J. Stat.*, 19(1):2780–2802, 2025. doi: 10.1214/25-ejs2398.

T. Lando, I. Arab, and P. E. Oliveira. Second-order stochastic comparisons of order statistics. *Statistics*, 55(3):561–579, 2021. doi: 10.1080/02331888.2021.1960527.

T. Lando, I. Arab, and P. E. Oliveira. Properties of increasing odds rate distributions with a statistical application. *J. Statist. Plann. Inference*, 221:313–325, 2022. doi: 10.1016/j.jspi.2022.05.004.

A. W. Marshall and I. Olkin. *Life distributions*. Springer Series in Statistics. Springer, New York, 2007. ISBN 978-0-387-20333-1.

A. Müller. Stochastic orders generated by integrals: a unified study. *Adv. in Appl. Probab.*, 29(2):414–428, 1997. doi: 10.2307/1428010.

T. Rychlik. Sharp bounds on variances of generalized order statistics. *Metrika*, 89(3): 355–371, 2026. doi: 10.1007/s00184-025-01013-2.

M. Shaked and J. G. Shanthikumar. *Stochastic orders*. Springer Series in Statistics. Springer, New York, 2007. ISBN 978-0-387-32915-4; 0-387-32915-3. doi: 10.1007/978-0-387-34675-5.

B. Wilfling. Lorenz ordering of power-function order statistics. *Statist. Probab. Lett.*, 30 (4):313–319, 1996. doi: 10.1016/S0167-7152(95)00234-0.